

Mahip Patel

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EDUCATION

Texas A&M University, Corpus Christi, TX, USA
Masters of Computer Science

May 2026
GPA: 3.6/4.0

Coursework: Computer vision, Advanced Computer Architecture, Artificial Intelligence, Advanced Operating System, Design and Analysis of Algorithms, Advanced Software Engineering

GSFC University, Vadodara, Gujarat, India
Bachelors of Computer Science and Engineering

May 2024
GPA: 7.5/10

WORK EXPERIENCE

Graduate Assistant & Peer Tutor, Texas A&M University, Corpus Christi, TX Aug 2025 – May 2026

- Supported 100+ students across two roles — as a **Graduate Assistant** at the Math Learning Center and as a Peer Tutor for algebra — managing tutoring schedules, documentation, and academic operations while delivering structured instruction across 2 class sections with a combined enrollment of **70 students**.
- Conducted regular office hours to provide individualized support, addressing student questions, clarifying course material one-on-one, and ensuring consistent academic progress throughout both semesters.

Web Scraping & Web Design (SWE Volunteer), Texas A&M University, Corpus Christi, TX May 2025 – Aug 2025

- Automated faculty research data collection using Python-based web scraping pipelines and improved internal university web-pages through layout, accessibility, and **UI/UX updates**.

Full Stack Software Developer(Part-Time), Sync Web Solutions, Vadodara, India Jan 2023 – Jul 2024

- Developed full-stack client-facing web applications using **React.js** and **MongoDB**, including responsive frontend components and database-driven features.
- Improved app usability, backend workflows, and deployment processes while supporting live production sites.

Full Stack Software Developer(Intern), Redspark, Vadodara, India Dec 2022 – Jan 2023

- Developed and maintained responsive web applications using **HTML, CSS, JavaScript, PHP, and MySQL**, improving user experience and application performance.
- Collaborated with senior developers to implement features, fix bugs, and optimize database performance.

Data Management Associate(Part-Time), Bhagwati Hardware, Vadodara, India Jan 2021 – Aug 2022

- Designed **SQL** and **Power BI** dashboards for pricing analysis and business reporting while cleaning operational datasets to improve accuracy and reduce reporting time.

PROJECTS

SmartAssist-Campus-Services-Assistant

- Built an AI-powered campus assistant using **Python, FastAPI**, and **MongoDB** with a custom RAG pipeline, enabling a dual-mode chatbot that handles campus FAQs, professor lookups, navigation, and student ticketing.
- Developed an **LLM-powered assignment checker** supporting PDF, DOCX, and TXT uploads with structural and grammar feedback, plus a community forum with peer discussions and an admin portal — deployed via **Docker**.

Online Food Ordering System

- Optimized backend workflows and database operations, improving application reliability and reducing response latency.
- Developed a full-stack food ordering platform using **Java, Servlets, JDBC, JSP, and HTML/CSS** with secure login, menu management, and real-time order tracking.

Resilient-Container-Security-Compliance-Engine

- Built the distributed cache module for a container security engine, implementing **SHA-256 layer hashing**, consistent hash ring routing across cache nodes, and a thread-safe LRU eviction policy — **achieving a 99.9% reduction** in redundant **Trivy scan** latency on repeated image deployments.
- Developed a fault-tolerant cache client with automatic node failover, write-through replication across all nodes, and dead node recovery, ensuring zero cache misses during ring rebalancing in a distributed multi-node environment.

Market Event Impact Tracker

- Developed a system to track and analyze real-time stock price reactions to global financial events using Alpha Vantage and news APIs; used **Kafka** to decouple data collection from anomaly detection, making the pipeline more reliable during spikes.
- Built serverless data ingestion pipelines using **Azure Functions** and **Azure Event Grid**, reducing compute overhead by **70%**.
- Applied Z-score-based anomaly detection to **1-min OHLC market data**, generating alerts for event-driven price spikes
- Correlated over **6 months** of S&P500 data with breaking news; achieved **95.6% acc** in identifying abnormal price behavior

SKILLS

Programming Languages

: Python, Java, JavaScript, PHP, .NET, SQL

Databases and Tools

: PostgreSQL, Redis, Kafka, MySQL, SQL Server, MongoDB

AI / Backend

: FastAPI, LLMs, RAG Pipelines, REST APIs / Web Services, Docker

Web / Frontend

: React.js, HTML/CSS, JSP, Servlets, JDBC

Data and Analytics

: NumPy, Pandas, Power BI, Excel, Tableau

Cloud and DevOps

: AWS, Microsoft Azure, Git, GitHub, Docker